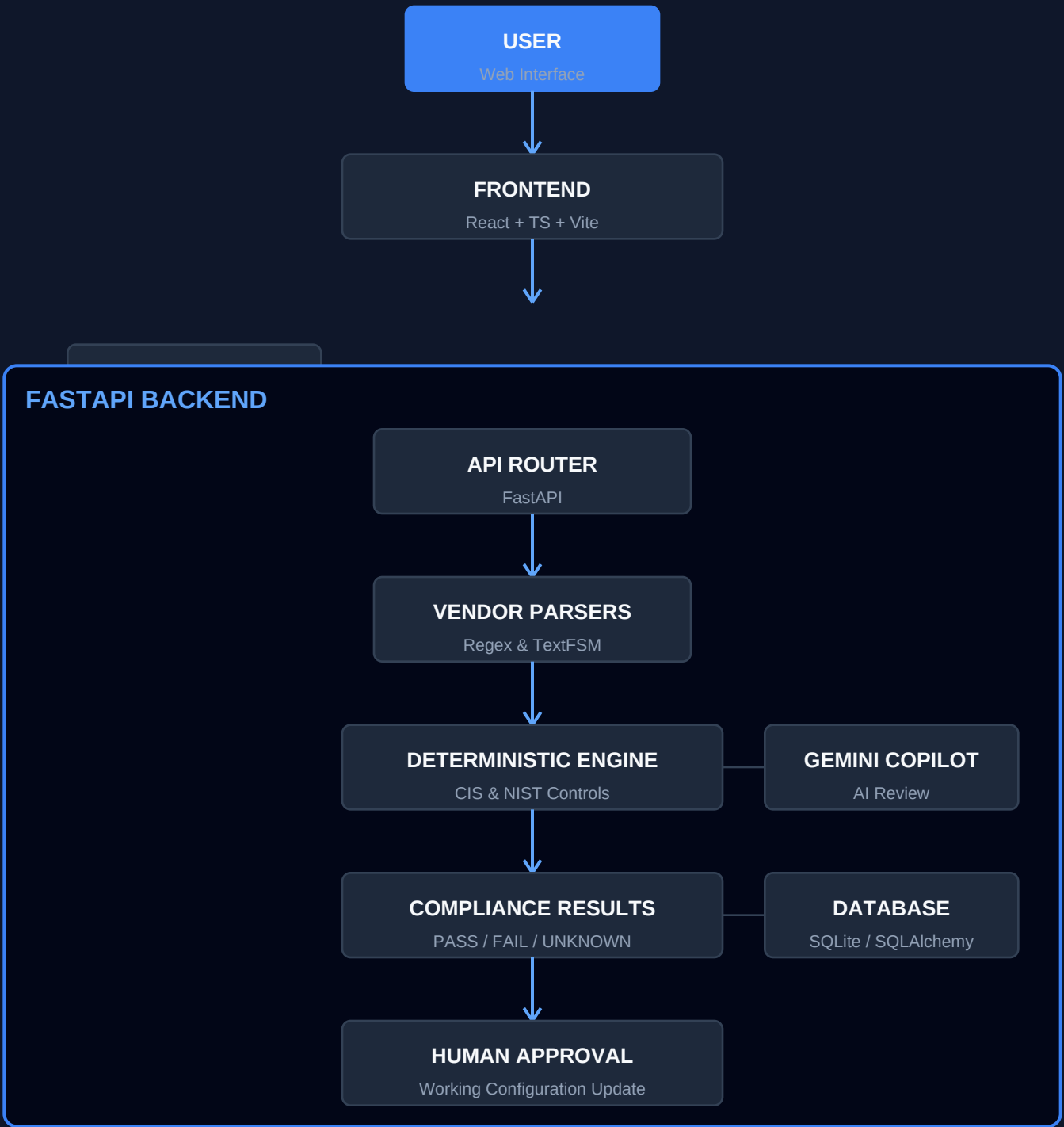


NETSECURE AI

SYSTEM ARCHITECTURE

AI-Driven Multi-Vendor Network Security Compliance Auditor | SIH26155

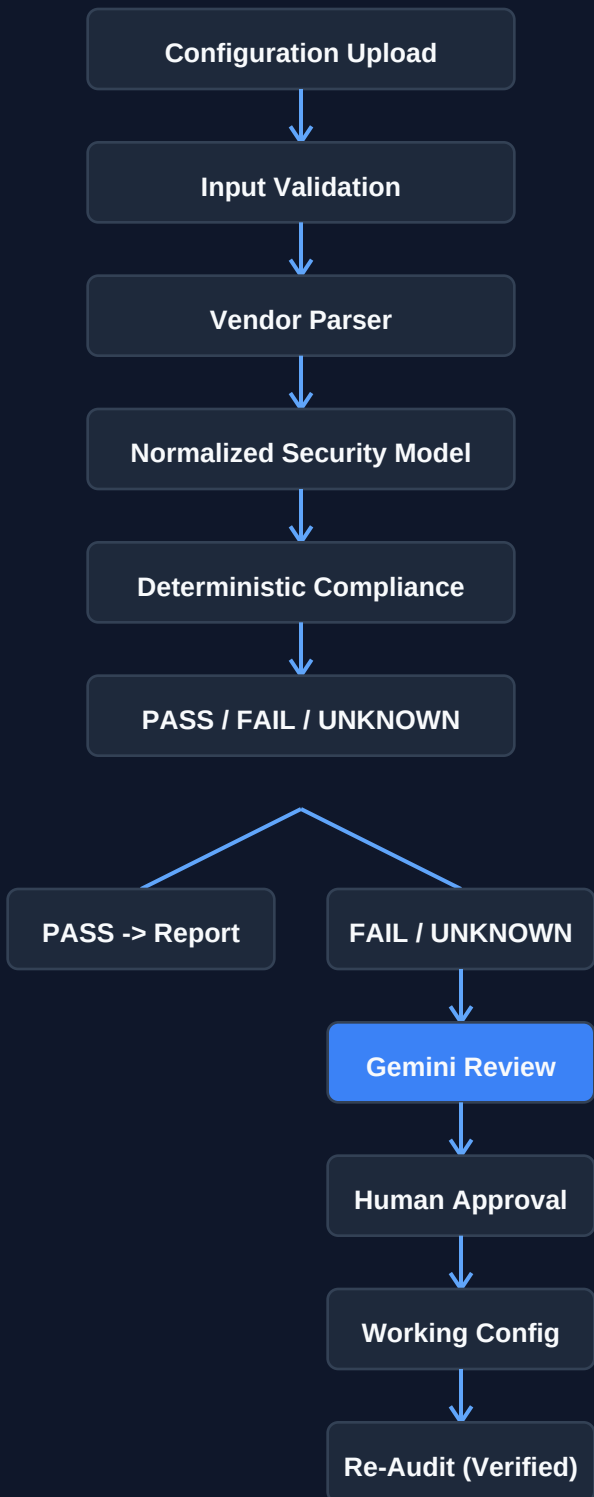


COMPONENT DESCRIPTION

- Frontend:** React + TypeScript + Vite provides the responsive enterprise security interface.
- Backend:** FastAPI handles authentication, auditing, compliance, AI orchestration and reporting.
- Parser Layer:** Vendor-specific parsers convert heterogeneous configurations into a normalized model.

DATA FLOW & SECURITY MODEL

1. AUDIT FLOW



2. SECURITY PRINCIPLES

Backend-only secrets

Gemini & Firebase Admin keys never enter frontend.

Authenticated APIs

Sensitive operations require Firebase auth tokens.

Human-controlled

AI does not directly execute arbitrary network commands.

Deterministic authority

Compliance decisions are based on strict rules.

Structured AI output

AI responses are validated via strict schemas.

Re-audit verification

Approved changes are deterministically re-evaluated.

3. CURRENT IMPLEMENTATION

Vendors: Cisco IOS, Fortinet, PAN-OS, Junos, Aruba, Check Point

Frameworks: CIS, NIST SP 800-53

AI: Google Gemini

Authentication: Firebase

4. FUTURE EXTENSION

STIG & ISO 27001 packs, Live device collection, Continuous compliance, Configuration drift detection, SIEM integration, Controlled live remediation.